

Frames and matrix designs

Sharp sampling complexity for Walsh restricted isometries

Difficulty: extreme **Importance:** broadly interesting

Status: Partially resolved

Rating rationale: Closing the uniform logarithmic sampling gap is a longstanding restricted-isometry barrier comparable in scale to the cyclic Fourier problem, with consequences for sparse recovery and fast sketches.

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Theorem 1 proves $m_*(2^d, k)/d \rightarrow C_k$ for $k = 2, 3, 4$, with $C_2 \approx 5.2988$, $C_3 \approx 16.7002$ and $C_4 \approx 36.3872$ (exact relative-entropy formulas are in Theorem 1); $k = 1$ needs one sample. Theorem 2 gives $m_*(N, k) \sim N \log N / h_+(1/2)$ for $k = N - o(N)$. Sampling is with replacement, distortion is $1/2$, and the results include success probability 0.9 . The uniform intermediate-sparsity target remains open.

See the [manuscript](#), [independent agent review](#), and [reproducible submission record](#). Independent agent review is not external human peer review or formal proof-assistant certification. Original problem, ratings and historical audits are retained below.

Let $N = 2^d$ with $d \geq 1$ and index rows and columns by $\mathbb{F}_2^d$. The normalized Walsh matrix is

$$(H_N)_{a,b} = N^{-1/2}(-1)^{a \cdot b}.$$

For $m \geq 1$, choose row indices $a_1, \dots, a_m$ independently and uniformly with replacement and set $\Phi = \sqrt{N/m}(H_N)_{(a_1, \dots, a_m), :}$. Let $E_{m,N,k}$ be the event that

$$\frac{1}{2}\|x\|_2^2 \leq \|\Phi x\|_2^2 \leq \frac{3}{2}\|x\|_2^2 \quad \text{for every } x \in \mathbb{R}^N \text{ with } |\text{supp } x| \leq k.$$

Define

$$m_*(N, k) = \min\{m \in \mathbb{N} : m \geq 1, \Pr(E_{m,N,k}) \geq 0.9\}.$$

Problem. Determine $m_*(N, k)$ up to universal multiplicative constants, uniformly for $1 \leq k \leq N$, including all necessary logarithmic factors. The same sample must work for all sparse vectors. This is a quantitative restatement of the published sampling gap, not a separately numbered conjecture; the success probability and distortion fix a convention.

The Walsh transform is the Fourier transform on $\mathbb{F}_2^d$. Its subspace structure differs from the cyclic Fourier ensemble in FR-02, so their lower bounds cannot be interchanged.

References

1. J. Błasiok, P. Lopatto, K. Luh, J. Marcinek, and S. Rao, *An improved lower bound for sparse reconstruction from subsampled Walsh matrices*, Discrete Analysis 2023:3, introduction and Theorem 3.1. [Paper](#).
2. I. Haviv and O. Regev, *The restricted isometry property of subsampled Fourier matrices*, Theorem 1.1 and its bounded-orthonormal-matrix scope. [Paper](#).

Status check — 2026-09-10

The lower-bound paper uses independent Bernoulli row inclusion; the upper-bound literature also treats independent draws with replacement, the explicit convention here. Its $\Omega(k \log k \log(N/k))$ obstruction applies in a specified intermediate sparsity range, not as an all-parameter formula. The logarithmic gap discussed there remains unclosed in the targeted search for later Walsh RIP results. End-point regimes must also be accounted for; in particular $m_*(N, 1) = 1$.

Audit update (2026-09-10): Rechecked the Walsh lower-bound record and Haviv–Regev upper bound, and searched for a later matching rate. The intermediate-sparsity obstruction is not an all-parameter formula. Difficulty is aligned with the comparable Fourier gap in FR-02. This is a bounded literature check, not a proof that no solution exists.